GGG-W-680a
April 7, 1964
SUPERSEDING
Fed. Spec. GGG-W-680
August 25, 1955

FEDERAL SPECIFICATION

WRENCHES, TAP AND REAMER, ADJUSTABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers adjustable wrenches used for holding and supplying torque to manually-rotated taps and reamers.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* Tap and reamer wrenches shall be of the following types and classes as specified (see 6.1):

Type I.—T-handle, adjustable jaws.
  Class 1.—Solid body.
  Class 2.—Ratcheting body.
Type II.—Straight handle, adjustable jaws.

1.2.2 *Sizes.* Sizes shall be as shown in applicable tables for each type and class as specified (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification:

Federal Specifications:

GGG-R-180—Reamers; Hand, Machine, Shell and Center; Arbors, Shell Reamer; and Pilots, Adjustable Hand Reamer.
GGG-T-70 — Tap, Thread Cutting (Standard, American National Form and Metric Spark Plug Thread, High-Speed Steel, Ground Thread, Hand).
GGG-T-581—Taps, Dies, Diestocks and Threading Set-Pipe Threading, Hand.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging Of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following publication forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D. C., 20402.)

3. REQUIREMENTS

3.1 **Preproduction sample.** When specified (see 6.1), before production is commenced, a sample of the item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 **Standard product.** Tap and reamer wrenches furnished under this specification shall be a standard product of the manufacturer except for such modifications from the manufacturer's standard product as may be specified herein.

3.3 **Illustrations.** The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of wrenches which are otherwise in accordance with the requirements of this specification.

3.4 **Material.** Unless otherwise specified, wrenches shall be of a good grade of steel suitable to attain the hardness requirements specified herein.

3.5 **Design and construction.** The wrenches shall be of the adjustable type for holding taps and reamers securely in the capacity ranges specified (see applicable tables). The wrench-holding capacities (see 4.4.2) are based on the sizes of squares on the shanks of hand taps, pipe taps and hand reamers specified in GGG-R-180, GGG-T-70, and GGG-T-581. Wrenches shall be of a solid body, ratcheting body, or straight handle type as specified herein.

3.5.1 *Type I wrench.* Type I wrench shall consist essentially of a body, T-handle, adjustable jaws and adjustable sleeve. The T-handle shall be utilized for applying torque to the wrench during the tapping and reaming operations. The adjusting sleeve shall be used to adjust the jaws for gripping and holding the taps and reamers. Type I, class 1 wrenches shall conform to table I (see figure 1); type I, class 2 wrench shall conform to table II (see figure 2).

TABLE I.—*Type I, class 1, sliding handle, solid body, T-handle, adjustable, tap and reamer wrench*

Size No.	Tap capacity range, minimum	Reamer capacity range, minimum	Wrench body length, minimum	Wrench handle length, minimum
	Inch	*Inch*	*Inches*	*Inches*
1	No. 0 to 1/4	3/0 to 1/4	2-1/2	3-1/4
2	1/4 to 1/2	1/4 to 3/8	3-3/8	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

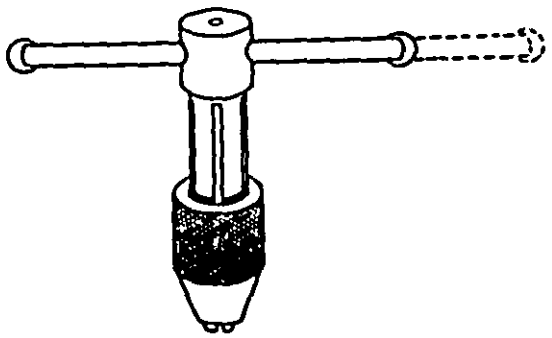


FIGURE 1.—Type I, class 1, wrenches.

TABLE II.—*Type I, class 2, ratcheting body, sliding handle, T-handle, adjustable, tap and reamer wrench*

Size No.	Tap capacity range, minimum	Reamer capacity range, minimum	Wrench body length, minimum	Wrench handle length, minimum
	Inch	*Inch*	*Inches*	*Inches*
1	No. 0 to 1/4	7/0 to 1/4	4	3-1/4

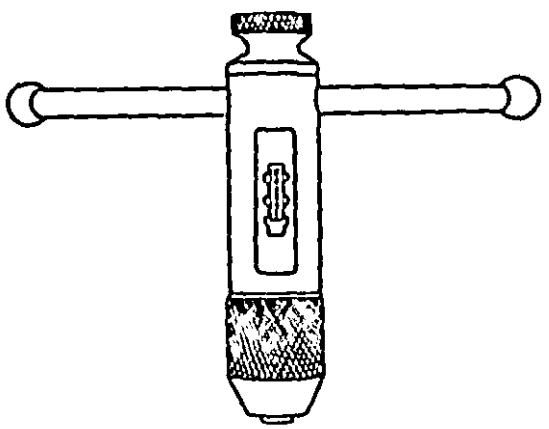


FIGURE 2.—Type I, class 2, wrench.

3.5.1.1 *Body.* The wrench body shall be fabricated from steel. The body shall include integral jaws or inserted jaws and shall be threaded for assembly of the adjusting sleeve. The top round section of the body shall be drilled 90 degrees to the vertical axis to accommodate the handle. The top surface of the body shall contain a 60 degree, countersunk center for use with machine centers. Class 1 wrenches shall consist of a solid body. Class 2 wrenches shall consist of a body inclosing a ratcheting device with a shifter capable of being set for three different positions; for clockwise ratcheting, for counterclockwise ratcheting, and for nonratcheting (ratchet mechanism locked). The ratcheting device and shifter shall operate smoothly.

3.5.1.2 *Jaws.* The jaws shall be either integral with the body or shall be of the inserted design. Inserted jaws shall be provided with springs which shall exert adequate pressure to hold the jaws firmly against the tapered bearing surface of the adjusting sleeve. Inserted jaws shall not become separated when the adjusting sleeve is removed. The jaws shall be hardened to Rockwell C50 to 58 (see 4.4.1).

3.5.1.3 *Adjusting sleeve.* The sleeve shall be threaded internally for assembly to the body. The inside sleeve surface shall have a hardened, tapered bearing surface which contacts the jaws. The outside surface shall be knurled. The sleeve shall be hardened to Rockwell C50 to 55 (see 4.4.1), except that the sleeve need not be hardened if the design is such that the jaws are affixed to the springs and are held within the sleeve by spring pressure.

3.5.1.4 *Handle.* The handle shall consist of a straight, round, steel rod and shall be assembled to the body to form a T-handle. The handle shall be adjustable and designed to be locked in the center position with both ends designed to prevent it from falling out, or it may be designed to be adjustable and held in any position by friction.

3.5.2 *Type II wrench.* Type II wrenches shall consist essentially of a body, handles, and jaws. The body, jaws, and handles shall be located about the same axis. The handles shall be utilized for applying torque to the wrench during the tapping and reaming operations. Type II wrenches shall conform to table III (see figures 3 and 4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

TABLE III.—*Type II, straight handle, adjustable tap and reamer wrench*

Size No.	Capacity—tap and reamer range			Over-all length	
	Bolt and screw taps, minimum	Pipe taps, minimum	Hand reamers, minimum	Minimum	Maximum
	Inches	*Inches*	*Inches*	*Inches*	*Inches*
0	No. 0 to 1/4		7/0 to 1/4	3	3-9/16
1	1/16 to 1/4		1/8 to 1/4	5	8
2	No. 8 to 1/2		11/64 to 3/8	8	12
4	No. 8 to 3/4	1/8 to 1/4	11/64 to 19/32	15	16
5	1/4 to 1-1/8	1/8 to 3/4	1/4 to 29/32	18	21
7	1/2 to 1	1/8 to 1/2	3/8 to 1/2	15	21
8	1/2 to 1-1/4	1/8 to 3/4	3/8 to 1	25	31
9	3/4 to 1-1/2	1/4 to 1	25/32 to 1-1/4	30	42
10	1 to 2-1/2	3/4 to 2-1/8	13/16 to 2-1/8	52	58
11	1-3/4 to 4	1-1/2 to 4	1-15/32 to 3	56	74



FIGURE 3.—Type II, size O, wrench.



FIGURE 4.—Type II, sizes 1 through 11, wrenches.

3.5.2.1 *Body.* The body shall be of drop forged steel, malleable iron, or steel bar stock.

3.5.2.2 *Jaws.* One jaw shall be fixed and the other shall be adjustable. For size 0, the fixed jaw shall be a part of the body. The adjustable jaw shall be integral with the adjusting screw with a full dog-point at the jaw end. For sizes 1 through 11, the fixed jaw shall be joined to or shall be a part of the body with the adjustable jaw joined to the rotating handle. The jaws shall be fabricated from steel and shall be hardened to Rockwell C50 to 65 (see 4.4.1).

3.5.2.3 *Handles.* Type II wrenches shall have two handles. For sizes 1 through 11 (see figure 4), one handle shall be integral with and screwed into the body; the other shall be designed to operate and adjust the movable jaw. The handles shall be fabricated of solid steel rod or steel tubing. The handles may be knurled or smooth. Size 0 wrench (see figure 3) shall have both handles integral with the body. One handle shall allow for assembly of an adjusting screw. The adjusting screw shall be knurled and shall have a 5/32-inch through hole for use with a steel rod or pin to tighten the jaws. Over-all length of size 0 wrench shall be measured when wrench is set at minimum capacity.

3.6 Threads. Threads shall be in accordance with Handbook H28—Screw Threads Standards for Federal Services.

3.7 Finish. The finish shall be in accordance with the manufacturer's commercial practice.

3.8 Marking for identification. The wrenches shall be stamped or otherwise permanently marked for identification with the manufacturer's name or a trademark of such known character that the source of manufacture may be readily determined.

3.9 Workmanship. The wrenches shall be clean with no evidence of rust or scale. Particular attention should be given to proper fit of mating parts and smooth operation of operating parts. All edges which are exposed to the operator's hands shall be beveled or rounded and all exposed surfaces shall be smooth (except for knurled surfaces) and free from burrs and sharp edges.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:42Z
Check the source to verify that this is the current version before use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required (see 3.1), it shall be examined for the defects listed in 4.3.3.1, dimensions specified herein, and tested in accordance with 4.4.2.

4.3 Sampling and inspection for acceptance. Sampling and inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 *Inspection of materials and components.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.3.2 *Intermediate inspection.*

4.3.2.1 *Hardness.* Jaws and adjusting sleeves shall be tested for hardness as specified in 4.4.1. The sample unit shall be one jaw or adjusting sleeve. A lot shall consist of jaws or adjusting sleeves of one type, class, and size that have been hardened. The inspection level shall be S-1 with an AQL of 1.5 defects, expressed in defects per hundred units.

4.3.3 *End product inspection.* The sample unit shall be one completely fabricated wrench. A lot shall consist of wrenches of one type, class, and size offered for acceptance at one time.

4.3.3.1 *Visual examination.* Each sample unit shall be visually examined for defects as set forth in table IV. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in defects per hundred units.

TABLE IV.—*Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Finish	Not in accordance with manufacturer's commercial practice.		X
	Not free from scale.		X
	Evidence of rust.	X	
Design and construction	Not type, class, and size specified.	X	
	Exterior surfaces not knurled where required.	X	
Type I	Jaws not integral with body; jaws not the inserted type on body; or jaws not held in sleeve by spring pressure (as applicable).	X	
	Body not threaded for the sleeve.	X	
	Countersink missing from top of body or not suitable for use with machine centers.	X	
	Jaws do not bear firmly against the inside bearing surface of the adjusting sleeve.	X	
	Inserted jaws not provided with springs or springs do not exert adequate pressure to hold jaws firmly against the tapered bearing sur-		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

TABLE IV.—*Classification of defects (Cont'd)*

Examine	Defect	Classification	
		Major	Minor
Design and construction (cont'd)			
Type I (cont'd)	face of the adjusting sleeve.	X	
	Inserted jaws become separated when the adjusting sleeve is removed.	X	
	Body and sleeve threads do not mate properly.	X	
	Body not solid (class 1).	X	
	Handle not sliding type and not designed to be locked or held in center position (class 2).	X	
	Handle falls out from body (class 2).	X	
	Ratcheting device missing or inoperative (class 2)	X	
	Shifter does not change direction of rotation (class 2).	X	
	Shifter cannot be set in three different positions as specified (class 2).	X	
Type II	Wrench does not have two jaws, one fixed and one adjustable.	X	
	Adjustable jaw not activated by rotation of handle.	X	
	Fixed jaw not part of body (size 0).	X	
	Wrench does not have one fixed and one adjustable handle (size 1 to 11).	X	
	Adjusting screw not designed as specified (size 0).	X	
	Adjustable screw does not operate smoothly.	X	
Workmanship	Exposed surfaces not smooth.		X
	Edges not beveled or rounded.		X
	Loose or not in proper alignment.	X	
	Part fractured, broken, malformed, or otherwise impaired.	X	
	Sharp burr or edges that may be injurious to personnel.	X	
	Functional part that requires abnormal force to operate.	X	
	Threads stripped, broken or impaired.	X	
Marking identification	Manufacturer's name or trademark missing, incorrect, illegible, or not permanent.		X

4.3.3.2 *Dimensional examination.* Examination shall be made of each sample unit to determine compliance with dimensional requirements specified herein. Any dimension that is not within the specified requirements shall be classified a defect. The inspection level shall be S-3 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

4.3.3.3 *Testing of the end product.* Each sample unit shall be tested as specified in 4.4.2. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

4.3.4 *Examination of preparation for delivery.* An examination shall be made to determine compliance with preservation and packaging, packing and marking requirements in accordance with section 5 and MIL-H-15424.

4.4 Tests.

4.4.1 The jaws and adjusting sleeves shall be tested for Rockwell hardness in accordance with method 243.1 of Fed. Test Method Std. No. 151 to determine conformance

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

to the requirements of 3.5.1.2, 3.5.1.3, and 3.5.2.2 as applicable.

4.4.2 *Capacity.* The wrenches shall be tested for tap and reamer holding capacity to determine conformance to the requirements of 3.5.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of preservation, packaging, and packing for civilian agency procurements.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A or C.* Tap and reamer wrenches shall be preserved and packaged in accordance with the applicable requirements of MIL-H-15424.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A, B, or C.* Tap and reamer wrenches, preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-H-15424.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, and size of wrench required (see 1.2 and 1.2.1).

(*c*) Whether preproduction sample is required (see 3.1).

(*d*) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Wrenches, not otherwise indexed by name, without attachments.
Carload minimum weight 30,000 pounds.

Motor:

Tools, hand, other than power.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.3 Supersession data. Changes in classification between this specification and GGG-W-680 are as follows:

Fed. Spec. GGG-W-680a
Type I, class 1
was
Fed. Spec. GGG-W-680
Type 1, class 1, style A and B

The following size wrenches covered in previous specification have been deleted:

Type I—class 2—Size 2
Type II—Sizes 3 and 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.

MILITARY CUSTODIANS:

Army—GL

Navy—Ships

Air Force—MOA

Interested activities:

Review

Navy—Ships

Air Force—MOA

User

Army—MO, WC, MU

Navy—Docks, MC

Preparing activity:

Army—GL

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:42Z
Check the source to verify that this is the current version before use.